

RESONANCE FREQUENCY

AC CIRCUITS

prepared by:

Gyro A. Madrona

Electronics Engineer

TOPIC OUTLINE

Series Resonance

Parallel Resonance

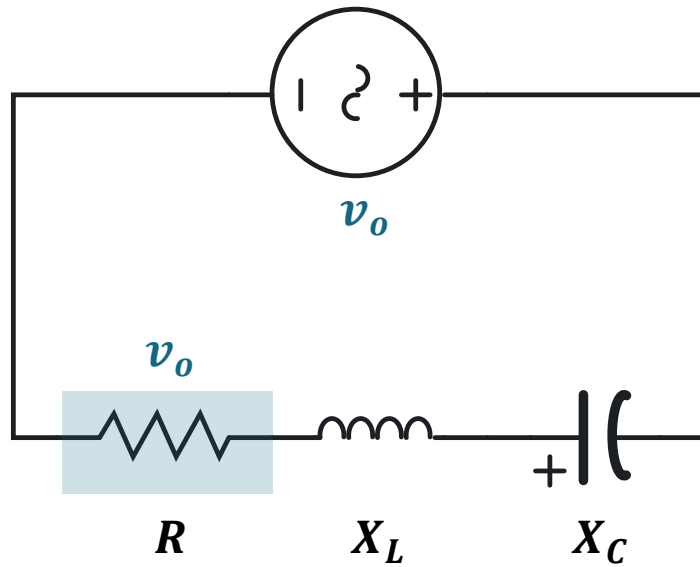


SERIES RESONANCE



SERIES RESONANCE

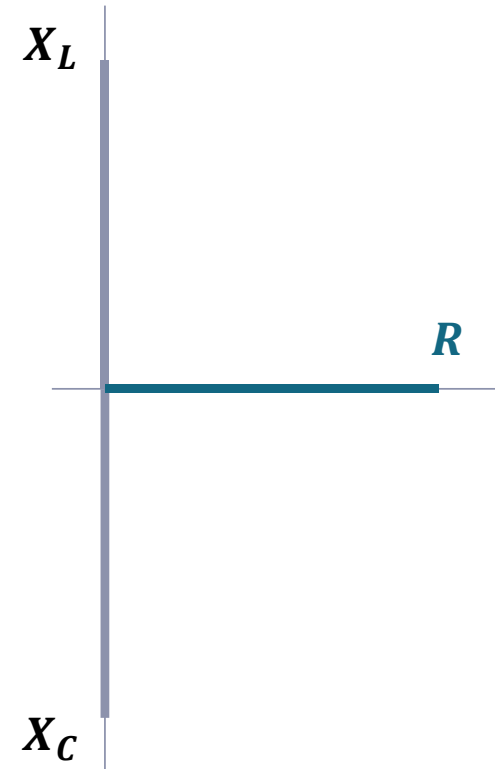
Circuit Diagram



Formula

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

Let $X_L = X_C$



EXERCISE

A series circuit consists of a 30Ω resistor, a 0.104H inductor, and a $40\mu\text{F}$ capacitor. If a variable-frequency 120V source is connected across its terminals, for the condition of resonant determine the following:

- The resonant frequency (f_R)
- The circuit current (i_R) and power (P_R)
- The voltage drop across the resistor (v_R), the inductor (v_L), and the capacitor (v_C)

Solution

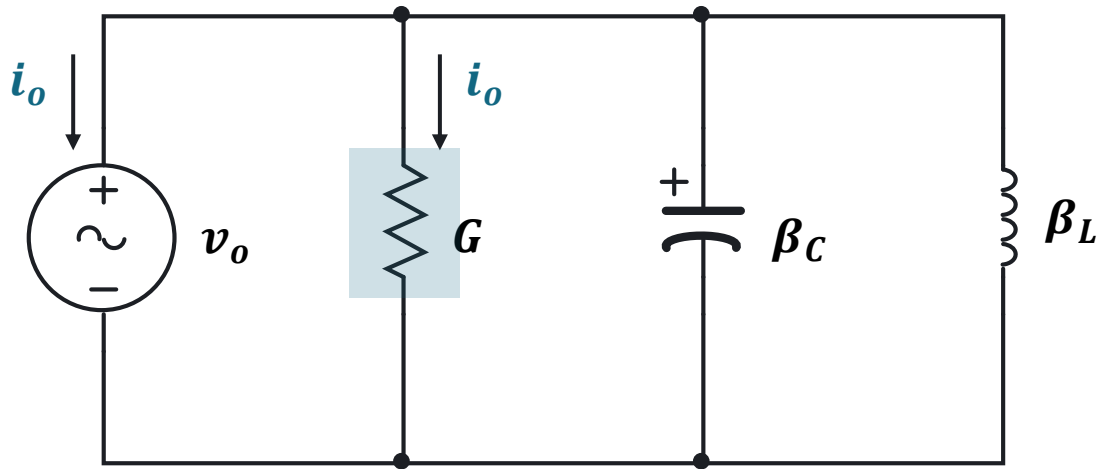


PARALLEL RESONANCE



PARALLEL RESONANCE

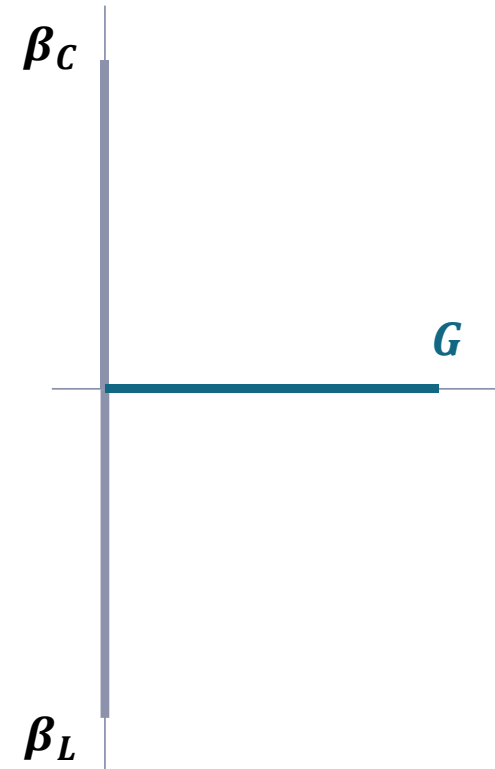
Circuit Diagram



Formula

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

Let $\beta_L = \beta_C$



EXERCISE

A parallel circuit consists of a 10Ω resistor, a 2H inductor, and a $50\mu\text{F}$ capacitor. If a variable-frequency 1V source is connected across its terminals, for the condition of resonant determine the following:

- The resonant frequency (f_R)
- The circuit current (i_R) and power (P_R)
- The voltage drop across the resistor (v_R), the inductor (v_L), and the capacitor (v_C)

Solution



RESONANCE FREQUENCY

$$f_R = \frac{1}{2\pi\sqrt{LC}}$$

Resultant reactance is zero [$X_L = X_C$]

Circuit behaves like a **pure resistance**

Total current is **in phase** with the impressed voltage

Power factor is **unity**



LABORATORY

